

LARYAK, Russian Federation

WMO: 238670

Lat: 61.100N

Lon: 80.250E

Elev: 186

StdP: 14.60

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.5	-36.5	-48.5	0.3	-43.1	-41.5	0.5	-36.0	14.9	4.2	13.8	5.5	2.2	240	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	84.6	66.0	81.5	65.0	77.7	63.5	68.6	79.9	66.5	76.8	64.9	74.6	5.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	92.7	72.9	62.9	86.5	70.8	61.0	80.8	69.6	33.0	79.3	31.4	77.2	30.1	74.6	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	13.6	11.9	-46.2	88.6	6.5	2.8	-50.9	90.6	-54.7	92.2	-58.3	93.8	-63.0	95.8		
(5)			-45.8	70.5	6.0	2.5	-50.2	72.4	-53.7	73.9	-57.1	75.3	-61.4	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.1	-8.6	-1.2	17.1	31.4	42.0	62.6	64.0	57.6	47.6	32.2	5.7	-3.2		
	DBStd	28.27	15.31	16.09	12.40	10.71	10.90	9.28	7.27	6.47	8.32	10.68	16.23	18.59		
	HDD50	8832	1816	1433	1020	560	303	11	0	9	142	558	1330	1650		
	HDD65	13281	2281	1853	1485	1008	715	154	110	243	521	1016	1780	2115		
	CDD50	1204	0	0	0	3	56	389	434	245	71	6	0	0		
	CDD65	179	0	0	0	0	3	83	78	14	1	0	0	0		
	CDH74	1615	0	0	0	0	41	793	652	126	3	0	0	0		
	CDH80	422	0	0	0	0	9	237	160	16	0	0	0	0		
	WSAvg	5.3	4.4	4.8	6.1	6.3	6.8	5.1	4.6	4.6	5.0	5.8	5.2	4.6		
	PrecAvg	22.6	1.1	0.9	1.1	1.3	1.9	2.5	2.9	3.1	2.2	2.3	1.8	1.5		
(15) Precipitation	PrecMax	29.8	2.7	1.7	2.3	2.8	5.2	5.7	6.0	6.6	4.6	4.4	3.5	2.9		
	PrecMin	15.5	0.2	0.2	0.2	0.1	0.5	0.3	0.7	0.1	0.3	1.1	0.9	0.7		
	PrecStd	4.3	0.5	0.4	0.5	0.7	0.9	1.1	1.5	1.5	1.0	0.7	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.6	34.1	42.3	62.2	80.6	88.9	88.5	81.4	73.2	61.7	39.1	31.2		
		MCWB	23.8	32.2	34.3	48.7	60.0	67.8	69.9	65.4	59.9	51.1	37.8	30.0		
	2%	DB	20.9	29.0	38.7	54.3	70.6	85.7	84.1	77.4	68.5	54.7	34.5	28.4		
		MCWB	19.8	27.8	32.6	44.3	55.5	65.5	66.5	62.8	57.6	48.1	33.3	27.1		
	5%	DB	16.9	23.8	35.8	50.1	65.5	82.8	81.2	74.2	64.5	49.0	31.7	25.0		
		MCWB	15.7	22.2	31.1	41.8	53.2	65.2	65.4	61.8	55.5	44.7	30.5	23.8		
	10%	DB	13.2	19.1	33.2	46.2	59.3	79.1	77.5	70.5	60.3	45.7	28.7	21.0		
		MCWB	12.3	17.7	29.8	39.5	49.9	63.6	64.2	60.5	53.4	42.4	27.5	19.8		
	0.4%	WB	23.8	32.5	35.9	49.7	60.9	71.0	72.1	67.3	61.3	52.9	37.9	30.2		
		MCDB	24.4	34.0	39.2	59.9	74.9	85.0	85.2	78.0	70.5	60.4	39.0	31.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	20.1	28.0	33.4	45.4	56.4	68.4	69.1	65.2	58.7	48.8	33.6	27.2		
		MCDB	21.0	29.1	37.6	53.4	67.9	81.6	79.6	73.6	66.1	54.0	34.3	28.4		
	5%	WB	15.8	22.1	31.8	42.5	53.7	66.3	67.3	63.3	56.6	45.4	30.5	23.9		
		MCDB	16.8	23.6	35.2	48.7	64.1	79.7	76.4	71.1	63.3	48.7	31.7	25.1		
	10%	WB	12.3	17.9	29.7	40.2	50.4	64.6	65.5	61.5	53.9	42.5	27.5	19.9		
		MCDB	13.1	19.0	33.2	45.4	58.5	76.2	73.6	68.9	59.5	45.1	28.8	21.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	14.8	17.0	16.7	15.4	17.1	16.7	15.8	13.3	9.4	12.8	12.8		
		MCDBR	15.1	11.9	15.0	18.7	22.2	23.3	22.0	22.2	19.8	14.3	8.6	14.3		
		MCWBR	14.5	10.7	11.4	12.2	11.7	9.0	8.5	10.0	10.6	9.6	7.5	13.8		
		MCDBR	14.4	11.8	14.8	17.0	21.7	21.5	19.3	19.2	17.8	13.4	7.3	14.6		
	5% WB	MCWBR	14.1	10.8	12.1	12.1	12.6	9.5	8.7	10.3	10.1	9.6	7.3	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.299	0.349	0.369	0.381	0.402	0.444	0.410	0.359	0.326	0.264	0.245		
	taud		1.953	2.004	1.955	2.053	2.285	2.328	2.181	2.268	2.369	2.243	2.003	1.895		
	Ebn at Noon		134	203	233	257	266	262	245	244	237	198	130	82		
	Edh at Noon		14	26	39	43	37	36	41	35	26	21	13	9		
(44) All-Sky Solar Radiation	RadAvg		123	408	900	1460	1603	1776	1607	1137	704	314	133	62		
	RadStd		11	34	55	141	163	128	182	150	89	40	30	5		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page